

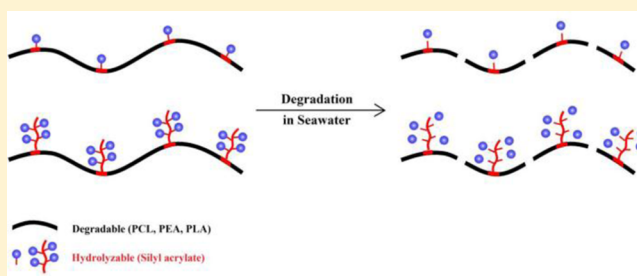
Degradable Polymers for Marine Antibiofouling: Optimizing Structure To Improve Performance

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S Supporting Information

ABSTRACT: Degradable polymers have proven effective for marine antibiofouling. It is important to improve their performance by optimizing their structures. We have prepared polyurethane with degradable polyester segments in the main chain and hydrolyzable poly(triisopropylsilyl acrylate) (PTSA) side chains and used it as the carrier of organic antifoulant (DCOIT). Our study demonstrates that the degradation rate of the polymer increases as the ester group density in the main chain or the length of side chain increases. On the other hand, the release rate of the organic antifoulant increases with the length of side chain. However, it shows a maximum as the ester group density increases. Marine field tests demonstrate that the antifouling of the system increases as the ester group density or the length of side chain increases.



INTRODUCTION

Marine biofouling has adverse effects on maritime industries.^{1,2} There is a great demand for environment-friendly antibiofouling systems since tributyltin (TBT) based coatings were banned.^{3–5} In the past years, poly(dimethylsiloxane) elastomers,^{6–9} protein resistant polymers,^{10–12} antimicrobial polymers,^{13,14} self-generated hydrogel,¹⁵ slippery surface¹⁶ based antifouling materials as well as screening environment-friendly antifoulants such as 4,5-dichloro-2-octyl-isothiazolone (DCOIT) and butenolide^{17–19} have been developed for this purpose. Actually, an effective system consists of polymer resins and antifoulants, where the controlled release of the antifoulants is critical for the long-term antifouling performance.

Polymer resins used in marine antibiofouling include soluble matrix (e.g., rosin), insoluble matrix (e.g., acrylic resin, chlorinated rubber) and self-polishing copolymers (SPCs).^{4,20,21} SPCs such as silyl acrylate, zinc or copper acrylates can generate a self-renewed surface, leading to the release of incorporated antifoulants.^{22,23} Even more, silyl acrylates can form a self-smoothing surface, which reduces the frictional resistance and offers unrivalled fuel saving. However, they exhibit poor marine antibiofouling performance on static conditions.²⁴ This is because the self-renewed ability of such polymers with a nonreactive main chain is determined by the hydrolysis of the side groups and the shear force generated by seawater flow. Moreover, the polymers do not degrade in marine environments, leading to ecological problems. Recently, we have developed degradable polymer based antibiofouling materials.^{25–27} Such polymers with degradable main chains can completely degrade into small molecules upon attacking of enzymes and water molecules in

marine environment. Particularly, the polymer coating can form self-polishing surface and exhibit antifouling ability on both dynamic and static conditions. However, optimization of polymer structures to control the release of antifoulants and regulate the antifouling duration is still the most important subject.

Polyurethanes have found a number of applications because of their good coating ability and adhesion to substrates. In this study, we have developed polyurethanes with different degradable polyester segments such as poly(ϵ -caprolactone), poly(ethylene adipate) or poly(L-lactide) and poly(silyl acrylate) side chains different in length. The degradation of the main chain and hydrolysis of side chains can be regulated by varying by the ester group density and the molecular weight of poly(silyl acrylate), respectively. Particularly, the polyurethane with degradable backbone can erode in marine environment without the help of shear force, which facilitates the antibiofouling in static conditions. Moreover, such polyurethane is not harmful to ecology because it completely degrades in a marine environment. We have investigated the degradation of the polymers, the release of the organic antifoulant (4,5-dichloro-2-octyl-isothiazolone) and the antibiofouling performance. We attempt to develop high performance marine antifouling systems applied to ships with low speed, offshore platform or aquaculture.

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Scheme 1. Synthesis of Polyurethane with Degradable Main Chain and Hydrolyzable Side Chains

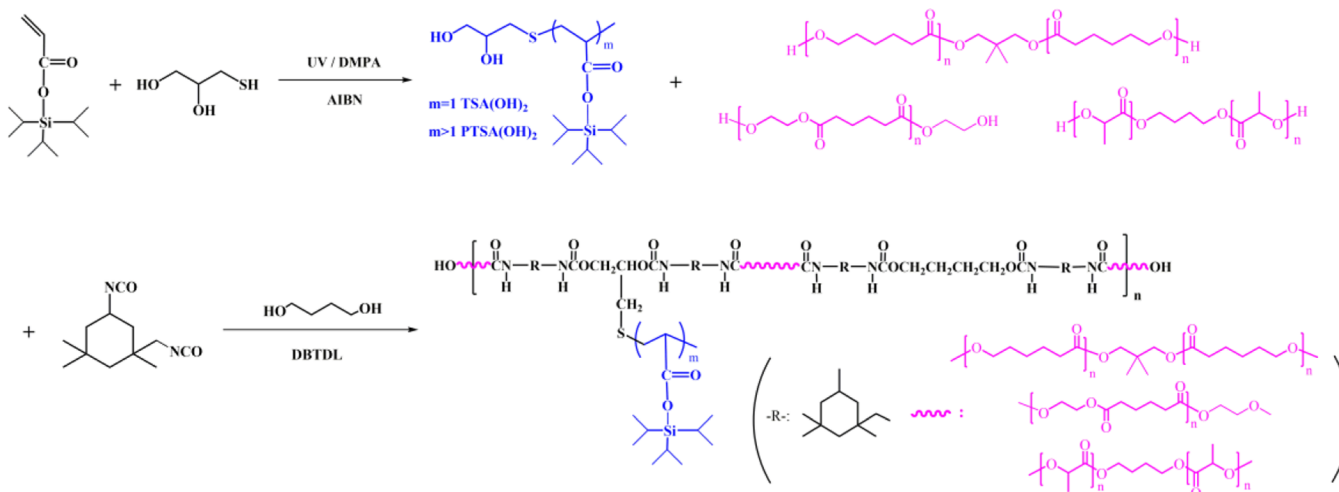


Table 1. Characterization Data of the Polyurethanes

sample ^a	SC content (wt %) ^b	($M_n \times 10^{-3}$) (g/mol) ^c	PDI ^c	γ_1 (MPa) ^d	γ_2 (MPa) ^e
PU-S336	4	41	1.4	1.76 $\pm$ 0.51	1.46 $\pm$ 0.10
PU-S1400	16	54	1.4	1.65 $\pm$ 0.18	1.38 $\pm$ 0.08
PU-S2600	30	42	1.5	1.03 $\pm$ 0.48	0.94 $\pm$ 0.12
PU-PCL-S336	16	39	1.5	1.72 $\pm$ 0.04	1.43 $\pm$ 0.15
PU-PEA-S336	16	54	1.4	4.36 $\pm$ 0.30	3.87 $\pm$ 0.34
PU-PLA-S336	16	43	1.6	1.09 $\pm$ 0.34	0.95 $\pm$ 0.10

^aFor PU-S336, PU-S1400 and PU-S2600, the feed weight ratios (PCL/IPDI/PTSA(OH)₂/BDO) are 1/6.33/0.46/4.87, 1/2.98/0.46/1.52 and 1/4.76/0.46/3.30, respectively. For PU-PCL-S336, PU-PEA-S336 and PU-PLA-S336, the feed weight ratios (Y/IPDI/TSA(OH)₂/BDO) are 1/5.19/1.90/2.29. ^bPTSA or TSA content determined by ¹H NMR. ^cDetermined by GPC. ^dAdhesion strength of polyurethane coating. ^eAdhesion strength of polyurethane coating with 10 wt % of DCOIT.

EXPERIMENTAL SECTION

Materials. Poly(ϵ -caprolactone) diol (PCL; M_w = 2000 g/mol) was from Perstorp. Poly(L-lactide) diol (PLA; M_w = 2000 g/mol) was from Daigang Biomaterial Co. (Shandong), poly(ethylene adipate) diol (PEA, M_w = 2000 g/mol) was from Yutian Chem. Co. (Shandong), Triisopropylsilyl acrylate (TSA) was synthesized following a procedure reported elsewhere.^{28,29} 3-Mercapto-1,2-propanediol (TG), 1,4-butanediol (BDO) as well as dibutyltin dilaurate (DBTDL) from Aldrich and isophorone diisocyanate (IPDI) from Aladdin were used as received. 2,2-Azobis (isobutyronitrile) (AIBN) was recrystallized twice from methanol. TSA was distilled under reduced pressure before use. PCL was dried under reduced pressure for 2 h prior to use. Tetrahydrofuran (THF) was refluxed over CaH₂ and distilled prior to use. 4,5-Dichloro-2-octyl-isothiazolone (DCOIT) was kindly presented by Thankful Chem. (Guangzhou). Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2013).³⁰ Other reagents were used as received.

Synthesis of Precursors. Dihydroxy-terminated triisopropylsilyl acrylate (TSA(OH)₂) was prepared via thiol-ene click reaction.^{31–33} Typically, 1.296 g of TG (12 mmol), 2.28 g of TSA (10 mmol) and 0.04272 g of DMPA (0.1667 mmol) were dissolved in 15.0 mL of THF. The solution was degassed with nitrogen for 20 min. The sample was incubated by irradiation with a UV-lamp (emitting nominally at 350 nm, light intensity 100%, 100 mW/cm²) for 1 h. The product was precipitated into deionized water twice, filtered and dried under vacuum at 40 °C for 24 h. The molecular weight of TSA(OH)₂ is

determined to be 336 g/mol from that of C₁₅H₃₂NaO₄SSi (HRMS m/z : 359.1683, Figure S1).

Dihydroxy-terminated poly(triisopropylsilyl acrylate) (PTSA(OH)₂) was synthesized following the procedure reported before.²⁶ By varying the molar ratio of TSA to TG, we prepared a series of PTSA(OH)₂ with different molecular weights (Figure S2). Two samples with molecular weight of 1400 and 2600 g/mol were used in this study.

Synthesis of Polyurethane with Degradable Main Chain and Hydrolyzable Side Chains. Polyurethane with TSA/PTSA side chains was synthesized via condensation reaction in THF under a nitrogen atmosphere (Scheme 1). IPDI reacted with PCL at 70 °C for 1 h, yielding a prepolymer. Subsequently, TSA(OH)₂ or PTSA(OH)₂ were introduced, and the reaction was conducted at 80 °C for another 1 h. Finally, BDO and DBTDL were added as the chain extender and catalyst, respectively, and the mixture was allowed to react at 80 °C for 3 h. The resulting polymer was precipitated in excess hexane under stirring and allowed to stand overnight. The precipitate was filtered, washed three times with hexane and vacuum-dried for 24 h. ¹H NMR (600 MHz, CDCl₃, ppm): 4.09 (CH₂CH₂CO), 4.06 (OCCH₂(CH₃)O), 4.05 (COCH₂CH₂CH₂CH₂CH₂O), 3.87 (OCH₂CH₂CH₂CH₂O), 2.33 (COCH₂CH₂CH₂CH₂CH₂O), 2.30 (COCH₂CH₂CH₂CH₂CH₂O), 1.69 (CH₂CH₂CO), 1.66 (COCH₂CH₂CH₂CH₂CO), 1.66 (OCCH₂(CH₃)O), 1.65 (COCH₂CH₂CH₂CH₂CH₂O), 1.52 (OCH₂CH₂CH₂CH₂O), 1.38 (COCH₂CH₂CH₂CH₂CH₂O), 1.27 (SiCH(CH₃)₂), 1.06 (NCH₂C(CH₃)CH₂C(CH₃)₂), 1.05 (SiCH(CH₃)₂). FTIR: 3360 (NH), 2950 (CH₃), 2860 (CH₂), 1730 (C=O).

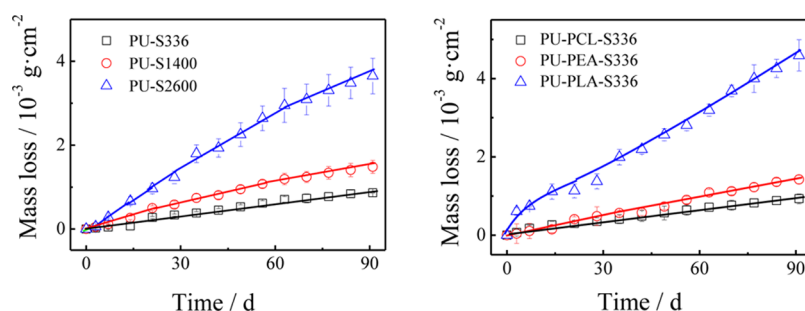


Figure 1. Time dependence of mass loss of the polyurethane in ASW at 25 °C.

For convenience, the polyurethane with TSA/PTSA side chains is designed as PU-S x , where x is the number-average molecular weight of TSA/PTSA, representing the length of side chain. The polyurethanes with PCL, PEA and PLA soft segments in main chains are designed as PU-Y-S336, where Y represents PCL, PEA or PLA degradable segment, and S336 represents the side chain (TSA) with molecular weight of 336 g/mol and content of 16 wt %. The characterization data are summarized in Table 1. The details can be found in Figure S3 and Figure S4.

■ CHARACTERIZATION

Proton Nuclear Magnetic Resonance Spectroscopy (^1H NMR). All ^1H NMR spectra were recorded on a Bruker AV600 NMR spectrometer with CDCl_3 as a solvent and tetramethylsilane (TMS) as an internal standard.

Fourier Transform Infrared Spectroscopy (FTIR). FTIR spectra were recorded on a Bruker VECTOR-22 IR spectrometer. The spectra were collected at 64 scans with a spectral resolution of 4 cm^{-1} by a KBr disk method.

High Resolution Mass Spectrometry (HRMS). HRMS was performed by using electrospray ionization (ESI) and recorded on Bruker maXis impact.

Gel Permeation Chromatography (GPC). The number-average molecular weights (M_n) and molecular weight distribution (M_w/M_n) were determined by GPC at 50 °C on a Waters 1515. A series of monodisperse polystyrenes were used as a standard, and THF (for PTSA(OH) $_2$) or DMF with LiBr (for polyurethanes) was used as the eluent at the flow rate of 1.0 mL/min.

Hydrolytic Degradation. The polyurethane film was prepared via a solution casting method. Typically, a solution of 30% (w/v) polyurethane in THF was dropped onto a glass fiber reinforced epoxy resin panel ($20 \times 20\text{ mm}$ in size). The panel coated with polyurethane was kept at room temperature for 7 days to remove the residual THF. Then, the weight (W_0) of each dried coating together with its panel was measured before dipping into a tank of ASW that was renewed every 2 weeks. After a certain period of time, the panel was taken out, rinsed with deionized water three times, gently dried with tissue before their weights ($W_{\text{t,wet}}$) were recorded and then dried at room temperature, and the weight ($W_{\text{t,dry}}$) of the panel was measured again. The mass loss was designated as $(W_{\text{t,dry}} - W_0)/\text{test area}$. The water adsorption was designated as $(W_{\text{t,wet}} - W_{\text{t,dry}})/\text{test area}$. For each sample, three coated panels were prepared and measured, and each data was obtained by averaging over three consistent measurements.

Contact Angle (CA) Measurement. The contact angle measurement was conducted on a Contact Angle System OCA40 (Dataphysics) at 25 °C by depositing a water drop of 3

μL on the polyurethane surface. The polyurethane film with a thickness of about $300\text{ }\mu\text{m}$ was prepared via solution casting. The samples were removed from ASW at different time intervals, rinsed using deionized water three times and dried at 25 °C before the CA measurements. For each sample, five different points were tested to obtain an average value.

Surface Roughness Measurements. BMT Expert 3D surface profile measurement system was used to measure the surface roughness and topography of the polyurethane film prepared via a solution-casting method with a thickness of approximately $300\text{ }\mu\text{m}$. For each sample, 10 line-scan measurements in the range of 2 mm were performed to obtain the average surface roughness (μm). The surface topography was observed by using area-scan measurement in the area of $0.5 \times 0.5\text{ mm}$.

Adhesion Test. The adhesion strength of polyurethane coatings on the epoxy resin panel was measured using Pull-Off Adhesion Tester (PosiTest AT-A Automatic) according to ASTM D4541-09.³⁴ The polyurethane films were prepared via a solution-casting method with a thickness of approximately $300\text{ }\mu\text{m}$. The measurement area is 20 mm in diameters and the pull rate was set at 0.2 MPa/s. For each sample, five different points were tested to obtain the average value. Considering that DCOIT is a small molecule with long alkyl chains (C8), its addition would decrease the mechanical properties of polyurethane. However, the decrease depends on its content. We measured the adhesion strength of the antifouling coatings containing 10 wt % of DCOIT, the drop in adhesion strength is less than 15% (Table 1).

Release Rate Test. DCOIT has been granted by European Union (EU) and US Environmental Protection Agency (EPA) as an antifouling products due to its rapid degradation in marine environments with half-life of $<24\text{ h}$ in seawater and $<1\text{ h}$ in sediment.³⁵ DCOIT release rate test was conducted under a static condition in ASW at 25 °C.^{36,37} The polyurethane matrix with 10 wt % DCOIT was prepared by solution mixing. Typically, polyurethane (1.08 g, 90 wt %) and DCOIT (0.12 g, 10 wt %) were dissolved in THF under stirring. The solution was coated on the epoxy resin panel ($40 \times 40\text{ mm}$) and kept at room temperature for 7 days before immersing in ASW. At a certain period of time, the panel was transferred from the holding tank to an individual measuring container. After immersion for 24 h, 100 mL of seawater was taken from 250 mL of seawater in the individual measuring container into a 250 mL glass separating funnel, and it was extracted with 10 mL of hexane. Then the concentration of DCOIT was determined quantitatively by an Agilent Cary 60 UV-vis spectrophotometer at wavelength 290 nm.³⁵ The release rate of DCOIT from the film was calculated with the following equation:

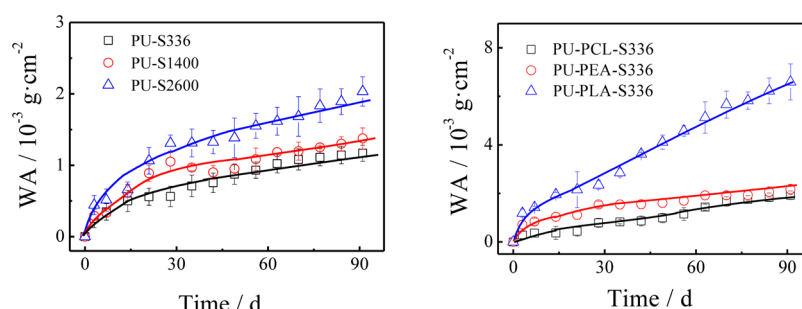


Figure 2. Time dependence of water adsorption (WA) of the polyurethane in ASW at 25 °C.

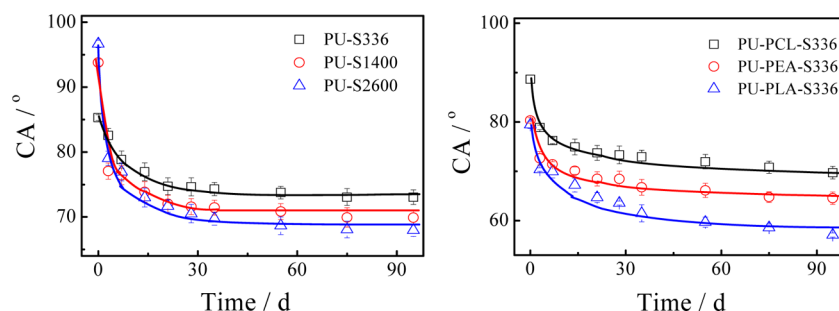


Figure 3. Time dependence of water contact angle (CA) on the polyurethane surface in ASW at 25 °C.

$$R = (C_{\text{DCOIT}} \times 250 \times D)/A$$

where R is the release rate of DCOIT ($\mu\text{g}/\text{cm}^2/\text{day}$), C_{DCOIT} is the DCOIT concentration in ASW ($\mu\text{g}/\text{mL}$), D is the immersion time in the individual measuring container (1 day) and A is the surface area of coated film (16 cm^2). DCOIT distribution in the cross section from the surface contacting ASW to the center of film was measured in terms of chlorine element by using a Quanta 200 (Philips-FEI) scanning electron microscope equipped with EDX analysis (SEM-EDX) with OXFORD INCA 350 system.³⁸ Each sample was sputter-coated with gold to minimize sample charging and operated at 20 kV.

Marine Field Tests. The field test was conducted at the inner Xiamen bay ($24^\circ 33' \text{N}$, $118^\circ 09' \text{E}$) in China according to the GB 5370-2007. Briefly, each sample coated on epoxy resin panel ($300 \times 80 \times 3 \text{ mm}$) was lowered into seawater at depths of 0.2–2.0 m from a stationary experimental raft on May 12, 2014. The coating consisted of polyurethane and DCOIT (10 wt %). After a certain time, the panels were taken out of the sea and carefully washed with seawater and photographed, and then they were immediately placed into the seawater to continue the test.

RESULTS AND DISCUSSION

Figure 1 shows the time dependence of mass loss of the polyurethane film in ASW. After 7 days of immersion, each film starts to lose weight. The mass loss is almost linearly related to time, indicating a constant degradation rate. For the polyurethanes with the same main chains but different side chains (PU-S336, PU-S1400 and PU-S2600), the hydrolytic degradation rate increases with the side chain length. Namely, the polyurethane with a longer side chain has a higher mass loss. This is understandable because a longer side chain possess a higher content of silyl ester group whose hydrolysis contributes to degradation.²⁶ For the polyurethanes with the same side chains but different main chains (PU-PCL-S336, PU-PEA-S336 and PU-PLA-S336), PU-PLA-S336 shows the highest mass loss

whereas PU-PCL-S336 has the smallest, and PU-PEA-S336 is in between. It is recognized that the hydrolysis of polyester is determined by ester group density.^{25,39} Because the ester group density increases in the order: PCL < PEA < PLA, the hydrolytic degradation rate is in the order: PU-PCL-S336 < PU-PEA-S336 < PU-PLA-S336. The self-polishing rate was estimated to be 0.01–0.08 $\text{mg}/\text{cm}^2/\text{d}$, which is comparable to that of degradable polyurethane with ϵ -caprolactone and glycolide (GA) copolyester oligomer segments (0.015–0.1 $\text{mg}/\text{cm}^2/\text{d}$).²⁵

Figure 2 shows water adsorption of the polyurethane immersed in ASW. For PU-S336, PU-S1400 and PU-S2600, the water adsorption increases with the side chain length because a longer side chain with a higher content of silyl ester groups generates more hydrophilic carboxylic acid groups after hydrolysis. For PU-PCL-S336, PU-PEA-S336 and PU-PLA-S336 with the same side chains but different main chains, the water adsorption is in the order: PU-PCL-S336 < PU-PEA-S336 < PU-PLA-S336 in that the hydrophilicity increases as the ester group density increases from PCL, PEA to PLA. Note that the large ($W_{\text{t,wet}} - W_{\text{t,dry}}$) for PU-PLA-S336 in the curves does not mean the polymer absorb much water. This is because $W_{\text{t,dry}}$ decreases with time due to the degradation whereas $W_{\text{t,wet}}$ related to the hydrolysis of silyl acrylate groups slightly varies. As a result, PU-PLA-S336 with the most mass loss ($W_0 - W_{\text{t,dry}}$) has the largest ($W_{\text{t,wet}} - W_{\text{t,dry}}$). We also determined the equilibrium water content (EWC) from $(W_{\text{t,wet}} - W_{\text{t,dry}})/W_{\text{t,wet}}$ to estimate the hydrophobicity of coating (see Figure S5), it exhibits the similar trend.

Figure 3 shows time dependence of water contact angle (CA) on the polyurethane surface. Before immersion, each PU-Sx sample has $\text{CA} > 85^\circ$. Moreover, CA increases with the side chain length. CA sharply decreases in 7 days but levels off after 30 days, indicating the hydrolysis of silyl ester groups. Note that CA values decrease with the length of side chain after hydrolysis. This is because more triisopropylsilyl ester groups hydrolyze as the side chain length increases, yielding more

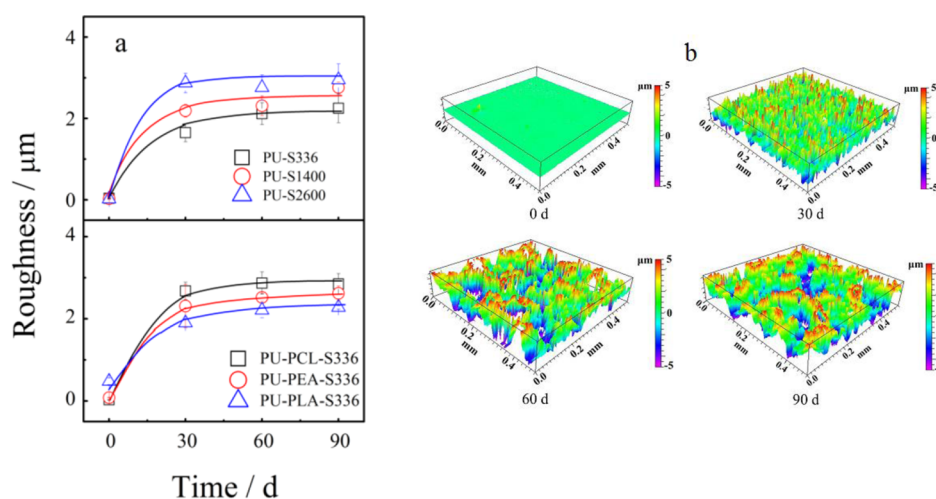


Figure 4. (a) Time dependence of surface roughness of the polyurethane coatings in ASW at 25 °C (b) Typical topography of the polyurethane coating (PU-S1400) in ASW at 25 °C as a function of time.

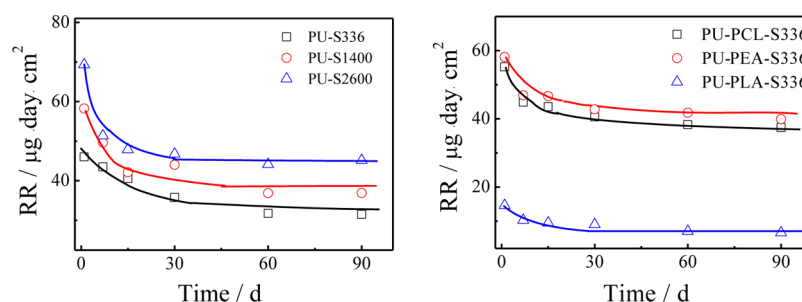


Figure 5. Time dependence of DCOIT release rate (RR) from the polyurethane coating in ASW at 25 °C.

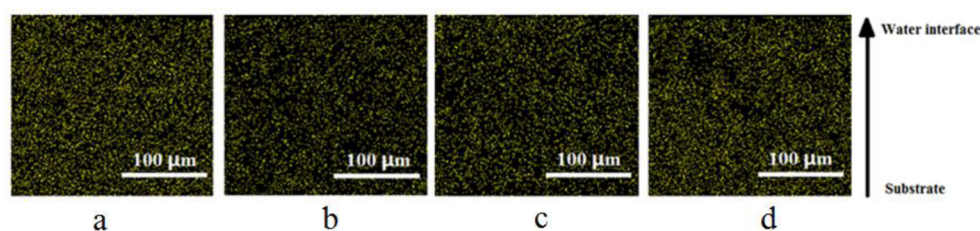


Figure 6. DCOIT distribution in the cross section of PU-S1400 as a function of immersion time in ASW at 25 °C detected by SEM-EDS: (a) 0 d, (b) 30 d (c) 60 d, (d) 90 d. Scale bar: 100 μm.

anionic carboxylate groups, so that the surface becomes more hydrophilic. For PU-PCL-S336, PU-PEA-S336 and PU-PLA-S336, CA always decreases with ester group density before and after the hydrolysis in that the surface hydrophilicity increases as the soft segment changes from PCL to PEA to PLA. Note that the surface roughness can also affect the CA. The large CA after prolonged water immersion should be due to the high surface roughness which probably offsets the CA decrease caused by hydrolysis. However, as shown in Figure 4, the surface roughness tends to level off after 30 days. Namely, the effect of surface roughness is constant.

Figure 4 shows time dependence of surface roughness of the polyurethane immersed in ASW at 25 °C. The surface roughness of PU-S_x increases from 0.03 to 2.5 μm in 30 days, indicating the occurrence of hydrolysis. After 60 days, the surface roughness tends to level off, indicating that PU-S_x exhibits self-smoothing in ASW with constant erosion rate (Figure 4b). The roughness increases with the side chain length because the content of silyl ester group increases. Anyhow, the

stable surface roughness after hydrolysis is favorable to ships in reducing resistance for ships in marine environments. For PU-PCL-S336, PU-PEA-S336 and PU-PLA-S336, the surface roughness increases with time within about 30 days and then levels off, similar to that of PU-S_x. This is the result of the hydrolysis of silyl ester group in the side chain and polyester segments in the main chain. On the other hand, the roughness decreases with the ester group density for a certain time. As we know, the swelling of hydrophilic moieties leads to the surface self-smoothing. As the ester group density increases, hydrophilic moieties are more uniformly distributed. Besides, a higher ester group density leads to a higher surface self-polishing rate, yielding a smoother surface. That is why the roughness decreases with ester group density.

Figure 5 shows the time dependence of DCOIT release rate in ASW at 25 °C. Each PU-S_x film has a high release rate in the early stage, but it gradually decreases and levels off after 30 days. Moreover, the release rate increases with the side chain length. Thus, we can regulate DCOIT release by varying the

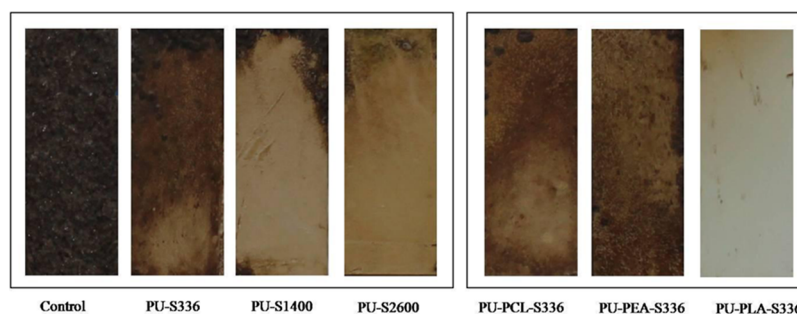


Figure 7. Marine antifouling performance of the polyurethanes with DCOIT (10 wt %) after 3 months of immersion in seawater.

side chain length. For the polyurethanes with the same side chains but different main chains, the release rate is in the order: PU-PEA-S336 > PU-PCL-S336 > PU-PLA-S336. Clearly, it is not consistent with that of degradation rate. PU-PEA-S336 exhibits the fastest release probably because it is poorly compatible with DCOIT, making DCOIT molecules be released readily. On the other hand, PU-PLA-S336 exhibits the slowest release of DCOIT though it has the highest density of ester groups. This is because PLA segment has high crystallinity with dense structure, which reduces the diffusion and release of DCOIT.^{40–42}

We examined DCOIT distribution in the polyurethane coating in ASW as a function of immersion time with chlorine element in DCOIT as the signal (Figure 6). In the typical sample PU-S1400, we can observe a constant diffusion layer and internal concentration of DCOIT from the surface contacting ASW to the center of film after 30 days or 90 days of immersion. Namely, DCOIT molecules are uniformly distributed in the polymer. The other samples show the similar trend (data not shown). Anyhow, SEM-EDS images together with the release rate data in Figure 5 demonstrate that DCOIT molecules are released from the coating at a constant rate. Accordingly, the polyurethane yields a controlled release of antifoulants.

The long-term antifouling performance of the system was evaluated by marine field tests. Figure 7 shows the typical images of panels coated with the polyurethane after immersion in seawater for 3 months. The control panel with epoxy was seriously fouled by marine organisms, indicating a high fouling pressure. For PU-Sx coatings containing DCOIT (Figure 7a), the fouling on the surface decreases as the length of side chain in the polyurethane increases, indicating that the antifouling ability increases. As discussed above, as the side chain length increases, the hydrolysis rate (self-polishing rate) and DCOIT release rate increase, so that the antifouling performance is improved. For PU-PCL-S336, PU-PEA-S336 and PU-PLA-S336 (Figure 7b), PU-PLA-S336 exhibits the best antifouling performance. In other words, the antifouling ability increases with the ester group density in the polyurethane. Note that PU-PLA-S336 has the slowest release of DCOIT. As discussed before,^{25–27} the antifouling is determined by both the self-polishing of the coating and the release of antifoulants. Here, the former should contribute more to the antifouling.

CONCLUSION

Degradable silyl acrylate based polyurethanes with different segments (poly(ϵ -caprolactone), poly(ethylene adipate) and poly(L-lactide)) in the main chain and different hydrolyzable side chain length have been prepared and used it as carrier of an

organic antifoulant. By varying the ester group density in the main chain and the side chain length, we can well regulate the erosion the polymer coating and control the release of antifoulant. As the side chain length increases or ester group density increases, the antifouling ability of the system consisting of the polyurethane and organic antifoulant increases. The system is expected to find applications in marine antibiofouling.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.6b02917.

HRMS for TSA(OH)₂, GPC curves for PTSA(OH)₂, ¹H NMR, FTIR spectra of the polyurethanes, time dependence of equilibrium water content (EWC) for PU-Y-S336 (PDF)

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Notes

The authors declare no competing financial interest.

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